

Problem 6.1, 6.4: Using the method of joints, determine the force in each member of the truss shown. State whether each member is in tension or compression.

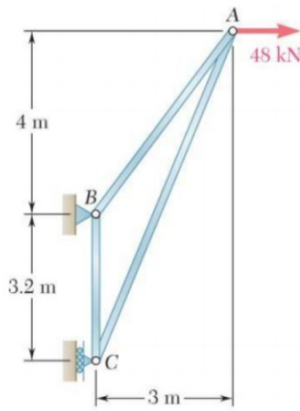
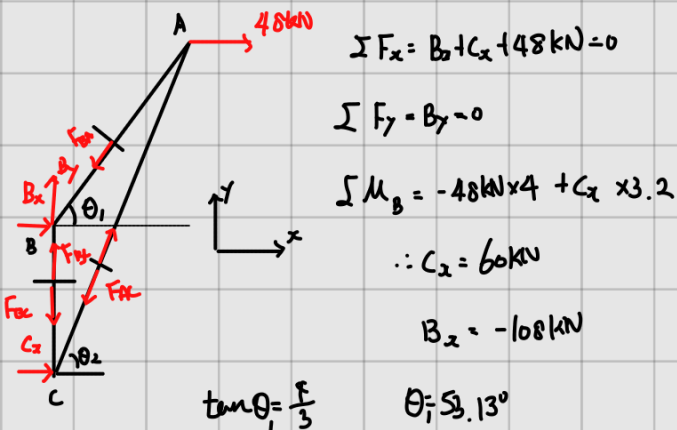


Fig. P6.1



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point B: $\underline{F_{BC}} + \underline{F_{BA}} + \underline{B_x} + \underline{B_y} = 0$

x-dir: $-F_{BA} \cos \theta_1 + B_x = 0$

$$\therefore F_{BA} = -180.0 \text{ kN}$$

$\therefore AB$ is in Tension

y-dir: $\underline{F_{BC}} + \underline{B_y} + \underline{F_{BA}} = 0$

$$F_{BC} + B_y - F_{BA} \sin \theta = 0$$

$$\therefore F_{BC} = F_{BA} \sin \theta = -144 \text{ kN}$$

$\therefore BC$ is in Tension

$$\theta_2 = \tan^{-1}(1.2/3) = 21.3^\circ$$

point C: $\underline{F_{BC}} + \underline{F_{AC}} + \underline{C_x} = 0$

x-dir: $-F_{AC} \cos \theta_2 + C_x = 0$

$$\therefore F_{AC} = 155.5 \text{ kN}$$

$\therefore AC$ is in Compression

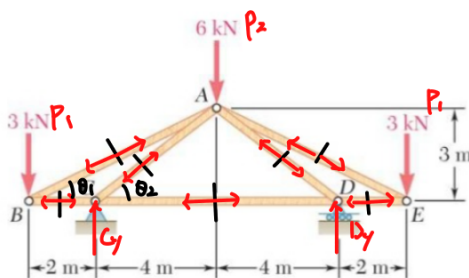


Fig. P6.4

point B: x-dir: $-F_{AB} \cos \theta_1 - F_{BC} = 0$

y-dir: $-F_{AB} \sin \theta_1 - P_1 = 0$

$$\therefore F_{AB} = -6.72 \text{ kN} \quad AB \text{ is in Tension}$$

$$F_{BC} = 6.01 \text{ kN} \quad BC \text{ is in Compression}$$

point C: x-dir: $F_{BC} - F_{AC} \cos \theta_2 - F_{CD} = 0$

y-dir: $-F_{AC} \sin \theta_2 + C_y = 0$

$$\therefore F_{AC} = 10.01 \text{ kN} \quad AC \text{ is in Compression}$$

$$F_{CD} = -2.00 \text{ kN} \quad CD \text{ is in Tension}$$

$$F_{AD} = 10.01 \text{ kN} \quad AD \text{ is in Compression}$$

$$F_{AE} = -1.72 \text{ kN} \quad AE \text{ is in Tension}$$

$$F_{DE} = 6.01 \text{ kN} \quad DE \text{ is in Compression}$$

대칭이므로

$$C_y = D_y$$

$$\therefore \sum F_y = -3 - 6 - 3 + C_y + D_y = 0$$

$$\therefore C_y = D_y = 6 \text{ kN}$$

$$\theta_1 = \tan^{-1}(3/6) = 26.5^\circ$$

$$\theta_2 = \tan^{-1}(3/4) = 36.8^\circ$$

대칭이므로

Problem 6.15: Determine the force in each of the members located to the left of line FGH for the studio roof truss shown. State whether each member is in tension or compression.

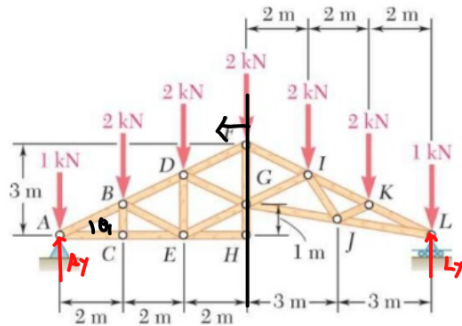


Fig. P6.15 and P6.16

$$\sum F_y = A_y + L_y - 12 = 0$$

$$A_y + L_y = 12$$

$$\sum M_A = -2 \times 2 - 4 \times 2 - 6 \times 2 - 8 \times 2 - 10 \times 2 - 12 \times 1 + 12 \times L_y = 0$$

$$L_y = 6 \text{ kN}$$

$$A_y = 6 \text{ kN}$$

$$\theta_1 = \tan^{-1}(1/2) = 26.5^\circ$$

$$\text{Point A: } x\text{-dir: } F_{AB} \cos \theta_1 - F_{AC} = 0$$

$$y\text{-dir: } -1 + A_y - F_{AB} \sin \theta_1 = 0$$

$$\therefore F_{AB} = 11.2 \text{ kN} \quad \text{AB is in comp}$$

$$F_{AC} = -10.02 \text{ kN} \quad \text{AC is in Tension}$$

$$\text{Point C: } x\text{-dir: } F_{AC} - F_{CE} = 0$$

$$F_{CE} = F_{AC} = -10.02 \text{ kN} \quad \text{CE is in tension}$$

$$y\text{-dir: } F_{BC} = 0$$

$$\text{Point B: } x\text{-dir: } F_{AB} \cos \theta_1 - F_{BD} \cos \theta_1 - F_{BE} \cos \theta_1 = 0$$

$$y\text{-dir: } F_{AB} \sin \theta_1 - F_{BD} \sin \theta_1 + F_{BE} \sin \theta_1 - 2 = 0$$

$$F_{BD} = 8.95 \text{ kN} \quad \text{BD is in compression}$$

$$F_{BE} = 2.24 \text{ kN} \quad \text{BE is in compression}$$

$$\text{Point H: } x\text{-dir: } F_{EH} = 0$$

$$y\text{-dir: } F_{GH} = 0$$

$$\text{Point E: } x\text{-dir: } F_{BE} \cos \theta_1 + F_{CE} - F_{EG} \cos \theta_1 - F_{EH} = 0$$

$$\therefore F_{EG} = -8.95 \text{ kN} \quad \text{EG is in Tension}$$

Problem 6.23: The portion of truss shown represents the upper part of a power transmission line tower. For the given loading, determine the force in each of the members located above HJ . State whether each member is in tension or compression.

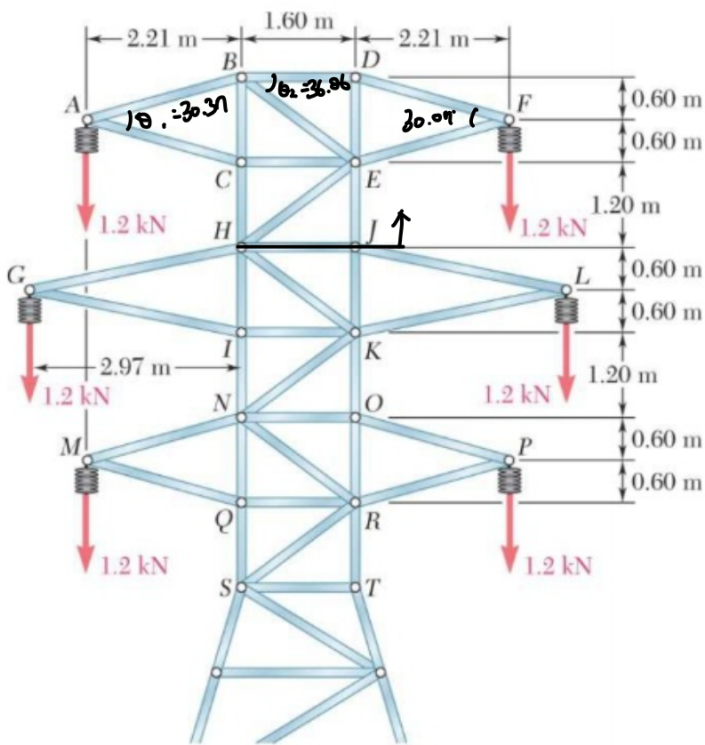


Fig. P6.23

$$\theta = \tan^{-1}(2.21/0.6) = 30.37^\circ \quad 2020/0804 \quad 0/32$$

Point A : $x\text{-dir} : -F_{AB} \cos \frac{\theta}{2} - F_{AC} \cos \frac{\theta}{2} = 0$

$$y\text{-dir} : -1.2 - F_{AB} \sin \frac{\theta}{2} + F_{AC} \sin \frac{\theta}{2} = 0$$

$$\therefore F_{AC} = 2.29 \text{ kN} \quad AC \text{ is in compression}$$

$$F_{AB} = -2.29 \text{ kN} \quad AB \text{ is in tension}$$

Point F : $F_{DF} = -2.29 \text{ kN} \quad DF \text{ is in tension}$

$$F_{EF} = 2.29 \text{ kN} \quad EF \text{ is in compression}$$

Point D : $x\text{-dir} : F_{BD} - F_{DF} \cos \frac{\theta}{2} = 0$

$$\therefore F_{BD} = -2.21 \text{ kN} \quad BD \text{ is in tension}$$

$$y\text{-dir} : F_{DE} + F_{DF} \sin \frac{\theta}{2} = 0$$

$$F_{DE} = 0.597 \text{ kN} \quad DE \text{ is in compression}$$

Point B : $x\text{-dir} : F_{AB} \cos \frac{\theta}{2} - F_{BE} \cos \theta_2 - F_{BD} = 0$

$$F_{BE} = 0$$

$$y\text{-dir} : F_{AB} \sin \frac{\theta}{2} + F_{BC} + F_{BE} \sin \theta_2 = 0$$

$$F_{BC} = 0.597 \text{ kN} \quad BC \text{ is in compression}$$

Point C : $x\text{-dir} : F_{AC} \cos \frac{\theta}{2} - F_{CE} = 0$

$$F_{CE} = 2.21 \text{ kN} \quad CE \text{ is in compression}$$

$$y\text{-dir} : -F_{BC} - F_{AC} \sin \frac{\theta}{2} + F_{CH} = 0$$

$$F_{CH} = 1.19 \text{ kN} \quad CH \text{ is in compression}$$

Point E : $x\text{-dir} : F_{BE} \cos \theta_2 + F_{CE} + F_{EH} \cos \theta_2 - F_{EF} \cos \frac{\theta}{2} = 0$

$$y\text{-dir} : -F_{BE} \sin \theta_2 + F_{EH} \sin \theta_2 - F_{DE} + F_{EJ} - F_{EF} \sin \frac{\theta}{2} = 0$$

$$\therefore F_{EH} = 0$$

$$F_{EJ} = 1.19 \text{ kN} \quad EJ \text{ is in compression}$$

Problem 6.45: A Warren bridge truss is loaded as shown. Determine the force in members CE, DE, and DF.

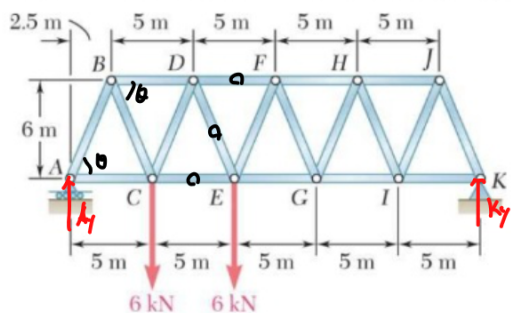


Fig. P6.45 and P6.46

$$\theta = \tan^{-1}(6/2.5) = 67.38^\circ$$

system/4

$$\Sigma F_y = A_y + K_y - 6 - 6 = 0$$

$$\Sigma M_A = -5 \times 6 - 10 \times 6 + 25 \times K_y = 0$$

$$K_y = 3.6 \text{ kN}$$

$$A_y = 8.4 \text{ kN}$$

Point A: x-dir: $+F_{AB} \cos \theta + F_{AC} = 0$

y-dir: $+F_{AB} \sin \theta + A_y = 0$

$\therefore F_{AB} = -9.1 \text{ kN}$ AB is in compression

$F_{AC} = 3.5 \text{ kN}$ AC is in Tension

Point B: x-dir: $-F_{AB} \cos \theta + F_{BC} \cos \theta + F_{BD} = 0$

y-dir: $-F_{AB} \sin \theta - F_{BC} \sin \theta = 0$

$F_{BC} = 9.1 \text{ kN}$ BC is in Tension

$F_{BD} = -7 \text{ kN}$ BD is in compression

Point C: x-dir: $-F_{AC} - F_{BC} \cos \theta + F_{CD} \cos \theta + F_{CE} = 0$

y-dir: $F_{BC} \sin \theta + F_{CD} \sin \theta - 6 = 0$

$\therefore F_{CD} = 13.5 \text{ kN}$ CD is in Tension

$F_{CE} = 1.8 \text{ kN}$ CE is in Tension

Point D: x-dir: $-F_{BD} - F_{CD} \cos \theta + F_{DE} \cos \theta + F_{DF} = 0$

y-dir: $-F_{CD} \sin \theta - F_{DE} \sin \theta = 0$

$\therefore F_{DE} = -13.5 \text{ kN}$ DE is in compression

$F_{DF} = 3.38 \text{ kN}$ DF is in Tension

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Problem 6.48: Determine the force in members GI , GJ , and HI .

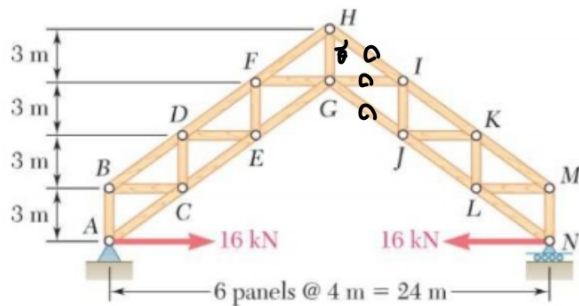
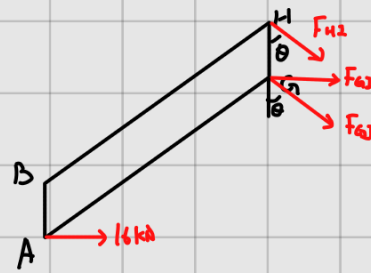


Fig. P6.47 and P6.48



$$\theta = \tan^{-1}(4/3) = 53.13^\circ$$

$$\sum F_x = 16 + F_{HI} \sin \theta + F_{GJ} + F_{FJ} \sin \theta = 0$$

$$\sum F_y = -F_{HI} \cos \theta - F_{FJ} \cos \theta = 0$$

$$\sum M_G = 9 \times 16 - 3 F_{HI} \sin \theta = 0$$

$$F_{HI} = 60 \text{ kN (Tension)}$$

$$F_{GJ} = -60 \text{ kN (Compression)}$$

$$F_{FJ} = -5 \text{ kN (Compression)}$$

Problem 6.51: A stadium roof truss is loaded as shown. Determine the force in members AB , AG , and FG .

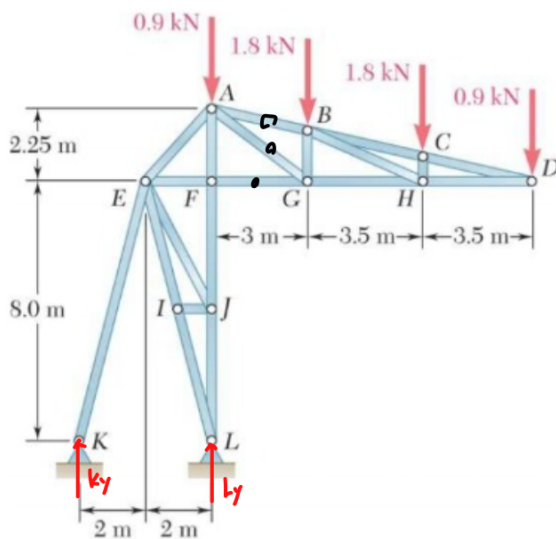


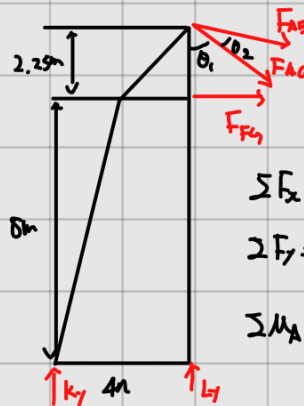
Fig. P6.51 and P6.52

$$\sum F_y = k_y + l_y - 0.9 - 1.8 - 1.8 - 0.9 = 0$$

$$\sum M_L = -4k_y - 3 \times 1.8 - 6.5 \times 1.8 - 10 \times 0.9 = 0$$

$$k_y = -6.525 \text{ kN}$$

$$l_y = 11.125 \text{ kN}$$



$$\theta_1 + \theta_2 = \tan^{-1}(10/2.25) = 77.32^\circ$$

$$\theta_1 = 53.13^\circ$$

$$\sum F_x = F_{AB} \sin(\theta_1 + \theta_2) + F_{AG} \sin \theta_1 + F_{FG} = 0$$

$$\sum F_y = -F_{AB} \cos(\theta_1 + \theta_2) - F_{AG} \cos \theta_1 + k_y + l_y = 0$$

$$\sum M_A = 2.25 \times F_{FG} - 4k_y = 0$$

$$\therefore F_{FG} = 41.6 \text{ kN (Compression)}$$

$$F_{AB} = 21.52 \text{ (Tension)}$$

$$F_{AG} = -14.07 \text{ (Compression)}$$

Problem 6.75: Determine the force in member BD and the component of the reaction at C .

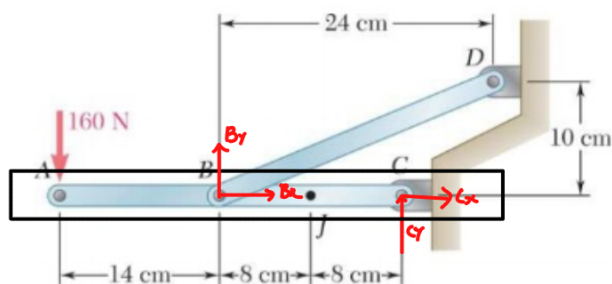


Fig. P6.75

$$\tan \theta = \frac{B_y}{B_x} = \frac{10}{24} \quad \theta = 20.200804 \text{ or } 20.2^\circ$$

$$\sum F_x = B_x + C_x = 0$$

$$\sum F_y = -160 + B_y + C_y = 0$$

$$\sum M_B = 160 \times 14 + 8 \times C_y = 0$$

$$\therefore C_y = -140 \text{ N}$$

$$B_y = 300 \text{ N}$$

$$B_x = 120 \text{ N}$$

$$C_x = -120 \text{ N}$$

$$\vec{F}_{BD} = 120 \text{ N} \hat{i} + 300 \text{ N} \hat{j} = 320 \text{ N} \angle 22.62^\circ$$

$$\vec{R}_C = (-120 \text{ N}) \hat{i} + (-140 \text{ N}) \hat{j} = 180.3 \text{ N} \angle 191^\circ$$

Problem 6.78: Determine the components of all forces acting on member ABD of the frame.

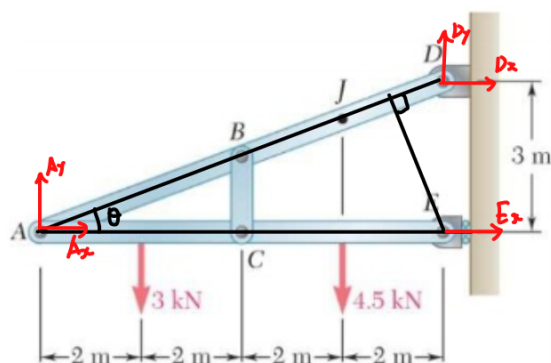


Fig. P6.78

$$\theta = \tan^{-1} \left(\frac{3}{6} \right) = 20.55^\circ$$

$$\sum F_x = D_x + E_x = 0$$

$$\sum F_y = D_y - 3 - 4.5 = 0$$

$$\sum M_D = 3 \times E_x + 2 \times 4.5 + 6 \times 3 = 0$$

$$\therefore E_x = -9 \text{ kN}$$

$$D_x = 9 \text{ kN}$$

$$D_y = 7.5 \text{ kN}$$

$$\vec{F}_D = 11.7 \text{ kN} \angle 20.55^\circ$$

$$\vec{F}_{BD} = 11.7 \text{ kN} \text{ (Tension)}$$

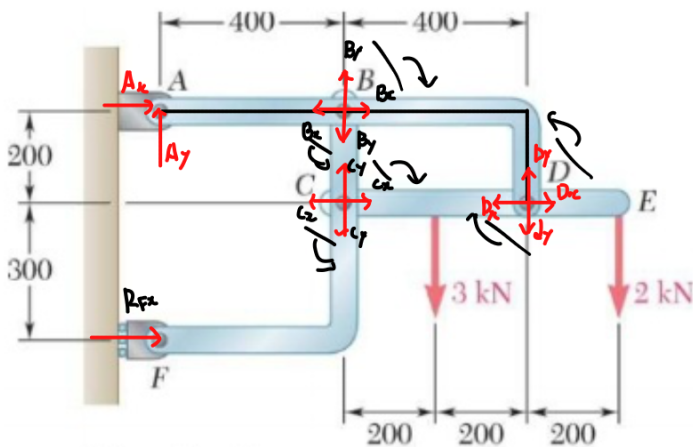
$$\text{Point A: } x\text{-dir } A_x + D_x = 0$$

$$A_x = -D_x = -9 \text{ kN}$$

$$y\text{-dir } A_y + D_y = 0$$

$$A_y = -D_y = -7.5 \text{ kN}$$

$$\vec{F}_A = 11.7 \text{ kN} \angle 200.55^\circ$$



Dimensions in mm

Fig. P6.105

중대 시스템이나

$$\Sigma F_x = A_x + R_{Fx} = 0$$

$$\Sigma F_y = Ay - 3 - 2 = 0$$

$$\sum \mu_A = -600 \times 3 - 1000 \times 2 + 500 \times R_{F_x} = 0$$

$$\therefore P_{F2} = 17.6 \text{ kN}$$

$$A_x = -7.6 \text{ kN}$$

$$A_y = 5 \text{ kN}$$

ABDmK

$$\sum F_x = A_x + B_x + D_x = 0 \quad 6k$$

$$\Sigma F_y = A_y + B_y + D_y = 0 \quad 6k$$

$$\Sigma M_B = -400A_y + 400D_y + 2600z = 0 \quad \text{OK}$$

$$\therefore R_D = (-7.6 \text{ kN})\hat{i} + (5 \text{ kN})\hat{j}$$

$$R_B = (-11.4 \text{ kN})\hat{L} + (-0.5 \text{ kN})\hat{J}$$

$$R_D = (19 \text{ kN})\hat{L} + (-4.5 \text{ kN})\hat{J}$$

Problem 6.107: Determine the reaction at F and the forces in members AE and BD .

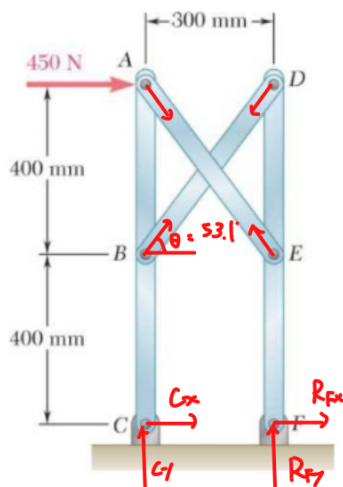


Fig. P6.107

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$$\sum F_x = 450 + C_x + R_{fx} = 0$$

$$\Sigma F_y = C_y + R_{Fy} = 0$$

$$\sum M_F = 300 \times 6 - 800 \times 4.5 = 0$$

$$G_y = 1200 \text{ N}$$

$$R_{Fy} = -1200 \text{ N}$$

DEF 5/4

$$\sum F_x = F_{BD} \cos \theta + F_{AE} \cos \theta + R_{F2} = 0$$

$$\sum F_y = F_{BD} \sin \theta - F_{AE} \sin \theta + R_{Fy} = 0$$

$$\sum M_E = -F_{BD} \times 400 + 400 R_{Dx} = 0$$

$$F_{B0} = 500 \text{ N}$$

$$I_{A1} = -1000 \text{ N}$$

$$R_{Ex} = 300 \text{ N}$$

$$\begin{aligned} \therefore R_F &= (300\text{N})\hat{i} + (-120\text{N})\hat{j} \\ F_{AE} &= 1600\text{N} \text{ (compression)} \\ F_{BD} &= 560\text{N} \text{ (Tension)} \end{aligned}$$

Problem 6.123: The press shown is used to emboss a small seal at *E*. Knowing that $P = 250$ N, determine (a) the vertical component of the force exerted on the seal, (b) the reaction at *A*.

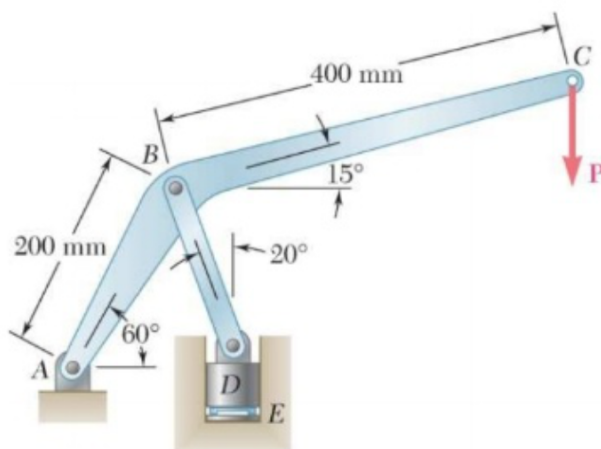
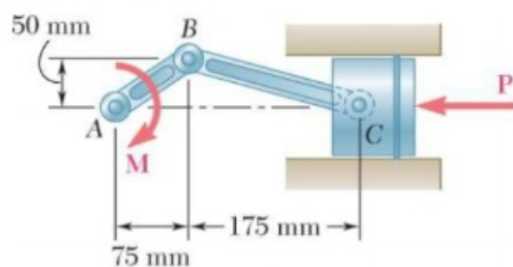
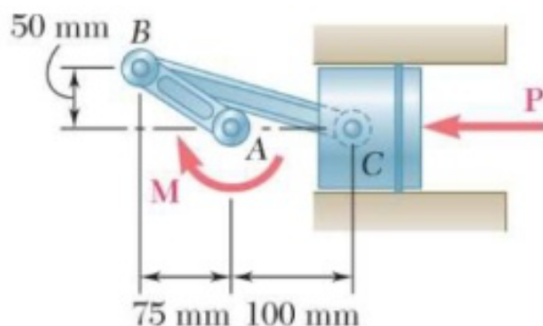


Fig. P6.123 and P6.124

Problem 6.131: A couple M of magnitude 1.5 kN·m is applied to the crank of the engine shown. For each of the two positions shown, determine the force P required to hold the system in equilibrium.

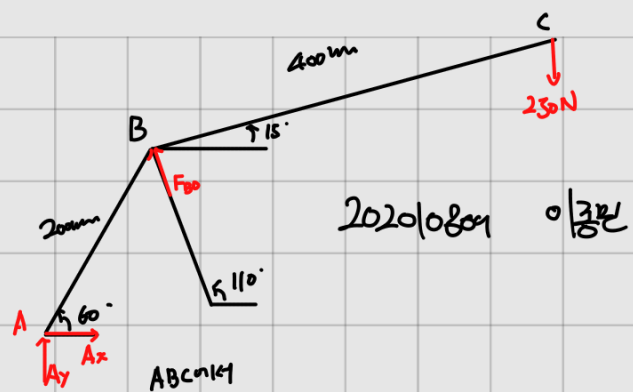


(a)



(b)

Fig. P6.131



$$\sum F_x = A_x - F_{BD} \sin 20^\circ = 0$$

$$\sum F_y = A_y + F_{BD} \cos 20^\circ - 250 = 0$$

$$\sum M_B = -200 \cos 60^\circ A_y + 200 \sin 60^\circ A_x - 400 \cos 15^\circ \times 250 = 0$$

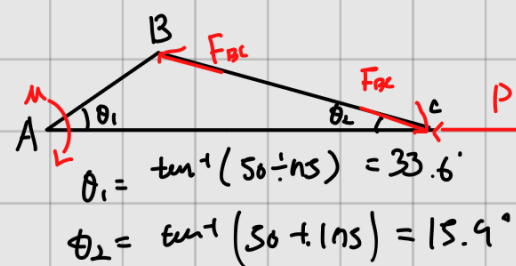
$$\therefore A_x = 211.4 \text{ N}$$

$$A_y = -495.7 \text{ N}$$

$$F_{BD} = 793.6 \text{ N}$$

$$\therefore a) F_{BD} \cos 20^\circ = 745.7 \text{ N}$$

$$b) R_A = (211.4 \text{ N})\hat{i} + (-495.7 \text{ N})\hat{j} = 565.1 \text{ N} \angle 296.9^\circ$$



$$\theta_1 = \tan^{-1}(50/175) = 33.6^\circ$$

$$\theta_2 = \tan^{-1}(50/175) = 15.9^\circ$$

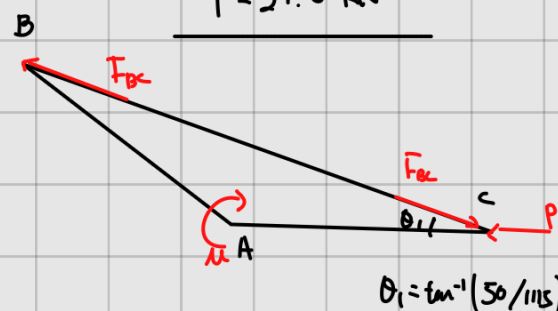
ABM

$$\sum M_A = -M + F_{BC} \sin \theta_2 \times 175 + F_{BC} \cos \theta_2 \times 50 = 0$$

$$\therefore F_{BC} = 21.8 \text{ kN}$$

$$\text{Point C: } \sum F_x = F_{BC} \cos \theta_2 - P = 0$$

$$P = 21.0 \text{ kN}$$



$$\theta_1 = \tan^{-1}(50/175)$$

$$= 15.94^\circ$$

ABM

$$\sum M_A = -M + F_{BC} \cos \theta_2 \times 50 - F_{BC} \sin \theta_2 \times 175 = 0$$

$$\therefore F_{BC} = 54.58 \text{ kN}$$

$$\text{Point C: } \sum F_x = F_{BC} \cos \theta_2 - P = 0$$

$$P = 52.48 \text{ kN}$$

Problem 6.145: Determine the magnitude of the gripping forces produced when two 300-N forces are applied as shown.

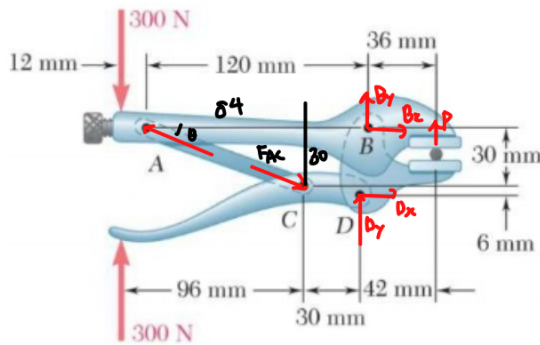


Fig. P6.145

$$\theta = \tan^{-1}(30 \div 89) = 19.65^\circ$$

$$\Sigma F_x = F_{AC} \cos \theta + D_x = 0$$

$$\sum F_y = -F_{AC} \sin 40^\circ + D_y + 300 = 0$$

$$\Sigma M_D = -6 F_{AC} \cos \theta + 30 F_{AC} \sin \theta - 126 \times 300 = 0$$

$$F_{AC} = 8518 \text{ N}$$

$AB \perp \angle 1$

$$\sum F_x = -F_{AL} \cos \theta + B_x = 0$$

$$\Sigma F_y = -300 + F_{AC} \sin \theta + B_y + P = 0$$

$$\sum M_B = 36P - F_{AC} \sin \theta \times 120 + 132 \times 300 = 0$$

$$\therefore P = 8448 \text{ N}$$